Project Planning Phase

Project Planning Template (Product Backlog, Sprint Planning, Stories, Story points)

Date	14 October 2026
Team ID	
Project Name	Metro Ticket Generating System
Maximum Marks	5 Marks

Product Backlog, Sprint Schedule, and Estimation (4 Marks)

Use the below template to create product backlog and sprint schedule

Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Priority	Team Members
Sprint-1	Project Setup & Data Architecture	USN-1	As an admin, I can set up the ServiceNow PDI and an update set to capture all changes.	1	High	
Sprint-1		USN-2	As an admin, I can create the Metro Station's Details table with the Station Name field.	2	High	
Sprint-1		USN-3	As an admin, I can add station records (Ameerpet, Madhapur, LB Nagar, Kukatpally, Jubilee Hills).	1	Medium	
Sprint-1	Catalog Form Setup	USN-4	As a passenger, I can open the 'Book A Metro Ticket' catalog item on the Service Portal.	2	High	
Sprint-1		USN-5	As a passenger, I can fill 8 form fields (stations, journey type, passengers, amounts, payment).	3	High	
Sprint-2	Fare Logic	USN-6	As a passenger, I can see the fare calculated automatically for my route and passengers.	5	High	
Sprint-2		USN-7	As a passenger, I can enter my payment mode when I choose Others (UI policy).	2	Medium	
Sprint-2	QR Code Integration	USN-8	As an admin, I can create the Metro QR Widget to show the QR image in a popup.	3	High	
Sprint-2		USN-9	As a passenger, I can see my QR ticket when I click Order Now and confirm the order.	5	High	
Sprint-3	Booking Storage & Automation	USN-10	As the operations team, I can see every booking in the Metro Ticket Booking table.	2	High	

Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Priority	Team Members
Sprint-3		USN-11	As an admin, I can build a flow that maps catalog variables to the booking table.	5	High	
Sprint-3		USN-12	As a station manager, I receive a catalog task to verify each new ticket.	2	Medium	
Sprint-4	Security & UAT	USN-13	As an admin, I can rely on default ACLs so that users cannot open the tables directly.	2	High	
Sprint-4		USN-14	As a tester, I can run UAT test cases on the portal (fare, visibility, QR, request).	3	High	
Sprint-4		USN-15	As a developer, I can fix defects found in testing (fare sys_ids, QR submission flag).	3	High	
Sprint-4	Documentation & Demo	USN-16	As a team, we can prepare the project documents and demo video.	2	Medium	

Project Tracker, Velocity & Burndown Chart: (4 Marks)

Sprint	Total Story Points	Duration	Sprint Start Date	Sprint End Date (Planned)	Story Points Completed (as on Planned End Date)	Sprint Release Date (Actual)
Sprint-1	9	2 Days	08 Oct 2026	09 Oct 2026	9	09 Oct 2026
Sprint-2	15	2 Days	10 Oct 2026	11 Oct 2026	15	11 Oct 2026
Sprint-3	9	2 Days	12 Oct 2026	13 Oct 2026	9	13 Oct 2026
Sprint-4	10	1 Day	14 Oct 2026	14 Oct 2026	10	14 Oct 2026
Total	43	7 Days	08 Oct 2026	14 Oct 2026	43	14 Oct 2026

Velocity:

For this project the team completed 43 story points in 4 sprints over 7 working days (08 – 14 October 2026). The team's average velocity (AV) is calculated as follows:

AV = Total story points completed ÷ Total sprint days = 43 ÷ 7 ≈ 6.1 story points per day

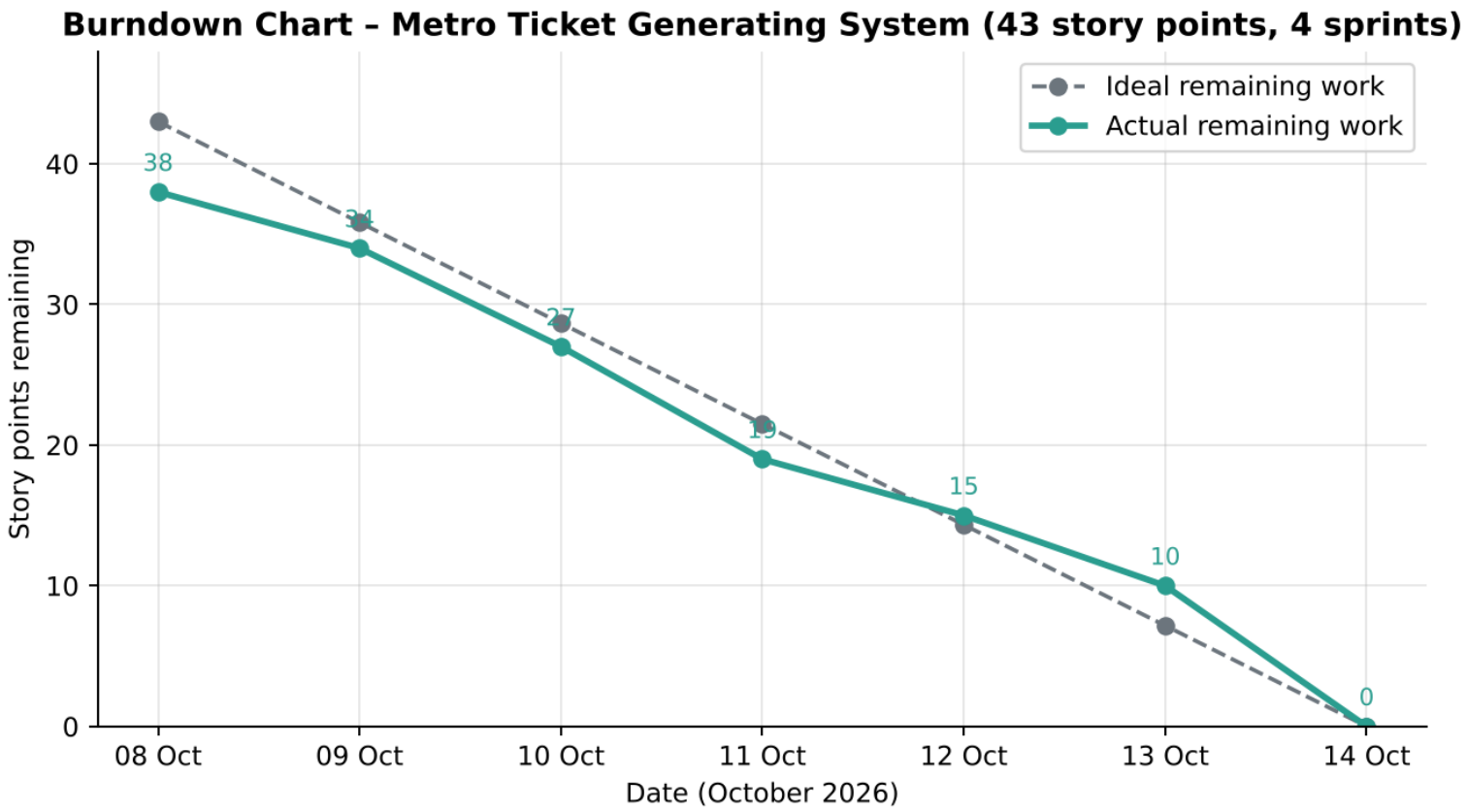
• Velocity per sprint = Total story points completed ÷ Number of sprints = 43 ÷ 4 = 10.75 story points per sprint

• Average velocity per day (AV) = Total story points ÷ Total sprint days = 43 ÷ 7 ≈ 6.1 story points per day

• Sprint-wise velocity: Sprint-1 = 9 ÷ 2 = 4.5 /day, Sprint-2 = 15 ÷ 2 = 7.5 /day, Sprint-3 = 9 ÷ 2 = 4.5 /day, Sprint-4 = 10 ÷ 1 = 10 /day

Burndown Chart:

A burn down chart is a graphical representation of work left to do versus time. It is often used in agile software development methodologies such as Scrum. However, burn down charts can be applied to any project containing measurable progress over time.
The burndown chart below shows the remaining story points for the Metro Ticket Generating System from 08 to 14 October 2026. The actual line stays close to the ideal line and reaches zero on 14 October 2026, when all 43 story points were completed.



https://www.visual-paradigm.com/scrum/scrum-burndown-chart/

https://www.atlassian.com/agile/tutorials/burndown-charts

Reference:

https://www.atlassian.com/agile/project-management

https://www.atlassian.com/agile/tutorials/how-to-do-scrum-with-jira-software

https://www.atlassian.com/agile/tutorials/epics

https://www.atlassian.com/agile/tutorials/sprints

https://www.atlassian.com/agile/project-management/estimation

https://www.atlassian.com/agile/tutorials/burndown-charts